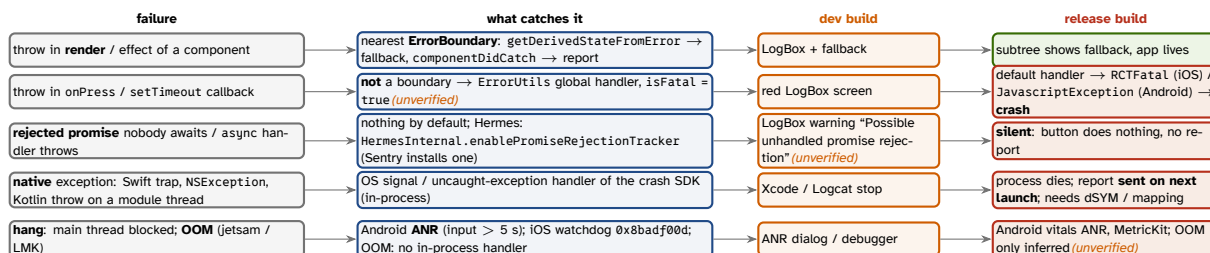


**In one line:** Every failure has **one catcher**: a **render** error goes to the nearest **error boundary**; an uncaught **JS exception** to ErrorUtils' **global handler** (fatal in release = crash); an **unhandled rejection** to nobody unless you install a tracker; a **native crash**, an **ANR** or **OOM** kills the process and is reported *next launch*. Design which errors the user sees, which retry, and which crash on purpose.



A busy **JS thread** is not an ANR: the main thread still answers the OS — the UI just stops reacting. Detect it yourself (JS FPS, a heartbeat), or users see a "frozen" app no tool reports.

## Error boundaries — what they catch

- A **class** component with static `getDerivedStateFromError(e)` (render the fallback) and `componentDidCatch(e, info)` (`info.componentStack` — minified in prod). **No hook equivalent** — use `react-error-boundary`.
- Catches errors thrown while React *renders, runs lifecycles/effects* of its children — and inside a `startTransition` function. **Not**: event handlers, `setTimeout/async` code, errors in the boundary itself.
- Route async errors in: `const { showBoundary } = useErrorBoundary()`, then `catch(e) { showBoundary(e) }`. Reset with `resetKeys` or a new key.
- Placement**: one at the root (last resort), one per **screen/route**, one around risky widgets (a chart, a 3rd-party view) — the rest of the screen keeps working.

## Example — global handler + rejection tracker (entry file)

```
const prev = ErrorUtils.getGlobalHandler();
ErrorUtils.setGlobalHandler((error, isFatal) => {
  report(error, { isFatal }); // persist first, send later
  if (!__DEV__ || !isFatal) return prev(error, isFatal); // LogBox
  showFatalScreen(error); // or prev(): the deliberate native crash
});
const H = (globalThis as any).HermesInternal;
if (H?.hasPromise?.()) H.enablePromiseRejectionTracker?({
  allRejections: true, // one tracker: chain others
  onUnhandled: (id: number, e: unknown) => report(e, { rejection: true }),
  onHandled: () => {},
});
```

- Chain, don't replace: crash SDKs (Sentry hooks `ErrorUtils`) and `LogBox` each rely on the handler — keep `prev` and call it, or one of them goes blind.
- A network send from a *fatal* handler may never leave before the process dies — persist, send next launch (*unverified*).
- Swallowing a fatal keeps a corrupted app alive; prefer a restart screen (`Updates.reloadAsync()` in Expo).

## Dev vs release

**LogBox** (default since 0.63; replaced `RedBox/YellowBox`): uncaught errors open full screen, logs show as notifications; `LogBox.ignoreLogs([...])` silences matches. None of it exists in release: a fatal is a crash, a rejection is silence. Test error paths in a **release build**.

## JS exception vs native crash vs ANR

|                | stack you get                             | to read it                                                               |
|----------------|-------------------------------------------|--------------------------------------------------------------------------|
| JS             | bundle offsets (Hermes: bytecode offsets) | JS <b>source map</b> — Hermes: packager map composed with the hermes map |
| native iOS     | addresses + <code>RCTFatal</code> frames  | <b>dSYM</b> matching the build UUID                                      |
| native Android | R8-obfuscated Java/Kotlin; NDK tombstone  | mapping.txt; native symbols                                              |
| ANR            | main-thread trace at the stall            | Play Console (Android vitals)                                            |

Upload per **build** and per **OTA update**; release + dist must name the exact bundle, or stacks point at wrong lines.

## Reporting — Sentry vs Crashlytics

- @sentry/react-native**: `Sentry.init({ dsn })`, `Sentry.wrap(App)`, `Sentry.ErrorBoundary`; JS errors (via `ErrorUtils`), rejections, native crashes, tracing; Metro/Xcode/Gradle steps upload maps (auth token in `sentry.properties`).
- @react-native-firebase/crashlytics**: native crashes automatic; JS via `crashlytics().recordError(e)` (non-fatal), `log()`, `setUserId()`. Either: scrub PII/tokens (`beforeSend`); tag user, update id, flags.

## What the user sees — expected vs unexpected

- Expected** (validation 4xx, offline, 404): errors as *values* — inline message, offline banner, disabled action. Never a crash screen.
- Transient** (5xx, timeout): backoff + jitter (TanStack Query retries queries 3×); idempotent calls or an idempotency key; then "Try again", not an endless spinner.
- Unexpected** (bug): boundary fallback with *retry* (reset) + report; fatal: report + restart.

## Traps

- ! "Boundaries catch everything" — not `onPress`, `async` or `native`.
- ! `async onPress` throws: no boundary, no crash, no report.
- ! One boundary around `<App/>`: a widget bug blanks the app.
- ! Great crash-free rate — JS freezes and OOMs never report.

## Remember

**Render** → **boundary** • **throw** → **global handler** • **promise** → **tracker** • **native** → **next launch**.